LAB EXPERIMENT

6. Perform basic Image Handling and processing operations on the image is to read an image in python and Convert an Image to erode using Erode function.

Code:

```
import cv2
import matplotlib.pyplot as plt
import numpy as np

# Read the input image
img = cv2.imread("scenary.jpg.jpg")

if img is None:
    print("Image not found. Check the file name!")
else:
    # Convert the image to grayscale
    gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

    # Create a binary image
    _, binary = cv2.threshold(
        gray, 127, 255, cv2.THRESH_BINARY
    )

    # Create a kernel
    kernel = np.ones((5, 5), np.uint8)

    # Apply erosion
    eroded = cv2.erode(binary, kernel, iterations=1)

    # Display the images
    plt.figure(figsize=(12, 4))

    plt.subplot(1, 3, 1)
    plt.imshow(gray, cmap="gray")
    plt.title("Original Grayscale Image")
    plt.axis("off")

    plt.subplot(1, 3, 2)
    plt.imshow(binary, cmap="gray")
    plt.title("Binary Image")
    plt.axis("off")

    plt.subplot(1, 3, 3)
    plt.imshow(eroded, cmap="gray")
    plt.title("Eroded Image")
    plt.axis("off")
```

```
plt.tight_layout()
plt.show()
```

